
UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 9: The Fast Fourier Transform (FFT)

Lecturer Prof. Justin Haldar

Collected by Gewenxin Yu

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1 Fast Fourier Transform (Cooley–Tukey)

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Notice: the DFT (Lecture 8) is

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk}, \quad 0 \leq k \leq N-1.$$

Direct DFT cost.

- For each k : N multiplies, $N-1$ additions.
- $\Rightarrow$ total: N^2 multiplies, $N^2 - N$ additions.

Suppose $x[n]$, $0 \leq n \leq N-1$, where $N = 2^M$ (**radix-2 FFT**). Let

$$\begin{aligned} y[n] &= x[2n], & 0 \leq n \leq \frac{N}{2} - 1 & \quad (\text{even-indexed samples}), \\ z[n] &= x[2n+1], & 0 \leq n \leq \frac{N}{2} - 1 & \quad (\text{odd-indexed samples}). \end{aligned}$$

¹cf. Mitra Secs. 11.3–11.5.

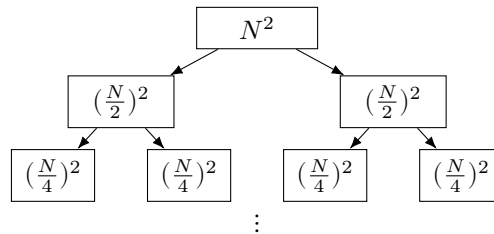
Claim.

We can easily compute $X[k]$ from the $\frac{N}{2}$ -point DFTs $Y[k]$ and $Z[k]$.

Splitting the problem reduces the computation:

$$\left(\frac{N}{2}\right)^2 + \left(\frac{N}{2}\right)^2 = \frac{N^2}{2} < N^2,$$

and each half can be split again, $\left(\frac{N}{2}\right)^2 \rightarrow \left(\frac{N}{4}\right)^2 + \left(\frac{N}{4}\right)^2 \rightarrow \dots$ (**divide-and-conquer**).



2 1.1 Historical Notes

- In 1994, the FFT was called “the most important numerical algorithm of our lifetime.”²
- Cooley and Tukey published their FFT paper in 1965. But many similar FFT algorithms exist from earlier authors — a “**rediscovery**.”
- The earliest known example is by **Gauss** in 1805.³

Motivating applications.

- **Gauss**: interpolate the trajectories of asteroids.
- **Tukey**: detecting Soviet nuclear tests using seismometers.

²G. Strang, “Wavelets,” *American Scientist*, vol. 82, 1994.

³M. T. Heideman, D. H. Johnson, and C. S. Burrus, “Gauss and the history of the fast Fourier transform,” *IEEE ASSP Magazine*, 1984.

3 2 Decimation-in-Time FFT (Radix-2)

Split the DFT sum into even ($n = 2p$) and odd ($n = 2p + 1$) indices:

$$\begin{aligned}
 X[k] &= \sum_{p=0}^{\frac{N}{2}-1} \left(x[2p] W_N^{2pk} + x[2p+1] W_N^{(2p+1)k} \right) \\
 &= \sum_{p=0}^{\frac{N}{2}-1} y[p] W_{N/2}^{pk} + \left[\sum_{p=0}^{\frac{N}{2}-1} z[p] W_{N/2}^{pk} \right] W_N^k \\
 &= Y[k] + Z[k] W_N^k, \quad 0 \leq k \leq \frac{N}{2} - 1,
 \end{aligned}$$

using $W_N^{2pk} = e^{-j\frac{2\pi}{N}2pk} = e^{-j\frac{2\pi}{N/2}pk} = W_{N/2}^{pk}$. For the upper half of the outputs,

$$\begin{aligned}
 X\left[k + \frac{N}{2}\right] &= \sum_{p=0}^{\frac{N}{2}-1} y[p] W_{N/2}^{(k+\frac{N}{2})p} + \left[\sum_{p=0}^{\frac{N}{2}-1} z[p] W_{N/2}^{(k+\frac{N}{2})p} \right] W_N^{k+\frac{N}{2}} \\
 &= Y[k] + Z[k] W_N^k W_N^{N/2},
 \end{aligned}$$

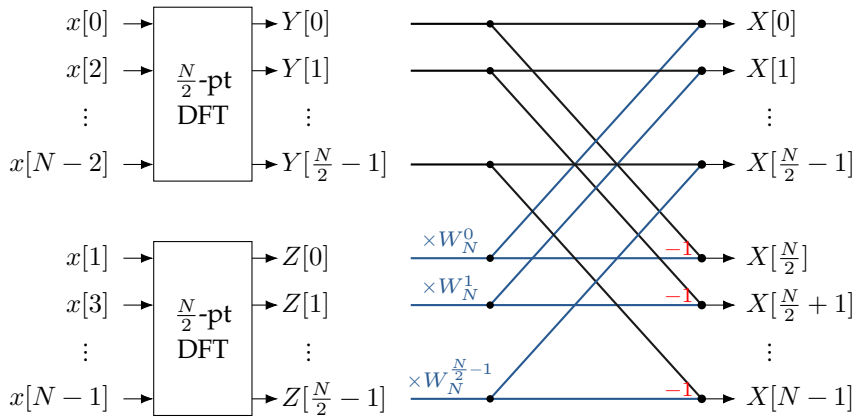
because

$$\begin{aligned}
 W_{N/2}^{N/2} &= \left(e^{-j\frac{4\pi}{N}} \right)^{N/2} = e^{-j2\pi} = 1 & (Y, Z \text{ are } \frac{N}{2}\text{-periodic}), \\
 W_N^{N/2} &= \left(e^{-j\frac{2\pi}{N}} \right)^{N/2} = e^{-j\pi} = -1.
 \end{aligned}$$

Radix-2 DIT butterfly.

$$\begin{aligned}
 X[k] &= Y[k] + W_N^k Z[k], \\
 X\left[k + \frac{N}{2}\right] &= Y[k] - W_N^k Z[k], \quad \text{for } 0 \leq k \leq \frac{N}{2} - 1.
 \end{aligned}$$

4 2.1 “Butterfly Diagram”



Each $Y[k]$ goes to both $X[k]$ and $X[k + \frac{N}{2}]$; $W_N^k Z[k]$ is added to $X[k]$ and subtracted (-1) from $X[k + \frac{N}{2}]$.

2.2 Cost of an N -point FFT

$$\text{Cost of an } N\text{-point FFT} = 2 \cdot (\text{cost of } \frac{N}{2}\text{-pt FFT}) + \begin{cases} N & \text{multiplies} \\ N & \text{additions} \end{cases}$$

If $N = 2^M \Rightarrow$ we can recurse:

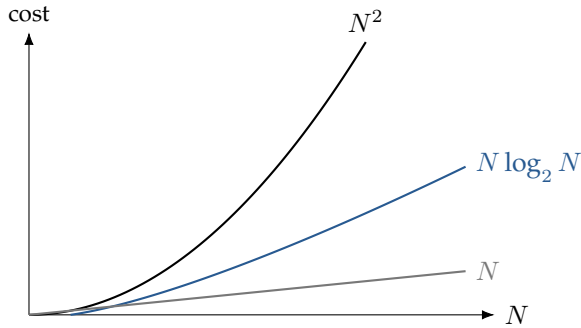
$$2 \cdot \left(2 \cdot (\text{cost of } \frac{N}{4}\text{-pt FFT}) + \begin{cases} \frac{N}{2} & \text{mult} \\ \frac{N}{2} & \text{add} \end{cases} \right) + \begin{cases} N & \text{mult} \\ N & \text{add} \end{cases}$$

$$\Rightarrow M = \log_2 N \text{ stages} \Rightarrow N \log_2 N \text{ multiplies / additions}.$$

Remark

(addendum to orig lecture note).

In the butterfly, the product $W_N^k Z[k]$ is computed *once* and then added to and subtracted from $Y[k]$; so each stage really needs only $\frac{N}{2}$ complex multiplications (and N complex additions), giving $\frac{N}{2} \log_2 N$ multiplications in total. Many of these are trivial ($W_N^0 = 1, W_N^{N/4} = -j$), so practical counts are lower still. Either way the order is $\mathcal{O}(N \log_2 N)$ versus $\mathcal{O}(N^2)$: for $N = 1024$, about 10^4 versus 10^6 .



3 Decimation-in-Frequency FFT

Now split the *time* range into its two halves:

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk} = \sum_{m=0}^{\frac{N}{2}-1} x[m] W_N^{mk} + \sum_{m=0}^{\frac{N}{2}-1} x\left[m + \frac{N}{2}\right] W_N^{\left(m + \frac{N}{2}\right)k}.$$

For k even, $k = 2q$:

$$\begin{aligned} X[2q] &= \sum_{m=0}^{\frac{N}{2}-1} \left(x[m] + x\left[m + \frac{N}{2}\right] \underbrace{W_N^{\frac{N}{2} \cdot 2q}}_{=1} \right) W_N^{2mq} \\ &= \sum_{m=0}^{\frac{N}{2}-1} \left(x[m] + x\left[m + \frac{N}{2}\right] \right) W_{N/2}^{mq}. \end{aligned}$$

For k odd, $k = 2q + 1$:

$$\begin{aligned} X[2q + 1] &= \sum_{m=0}^{\frac{N}{2}-1} \left[x[m] + x\left[m + \frac{N}{2}\right] \underbrace{W_N^{\frac{N}{2}(2q+1)}}_{=-1} \right] W_N^{(2q+1)m} \\ &= \sum_{m=0}^{\frac{N}{2}-1} \left[x[m] - x\left[m + \frac{N}{2}\right] \right] W_{N/2}^{mq} W_N^m. \end{aligned}$$

DIF in words.

The even-indexed outputs are the $\frac{N}{2}$ -point DFT of $x[m] + x[m + \frac{N}{2}]$; the odd-indexed outputs are the $\frac{N}{2}$ -point DFT of $(x[m] - x[m + \frac{N}{2}])W_N^m$. The butterfly (add/subtract, then twiddle) now comes *before* the smaller DFTs, and the outputs come out in even/odd order.

6 4 Beyond Radix-2

- Can generalize to **radix-3** ($N = 3^M$) $\Rightarrow$ perform 3 DFTs of size $\frac{N}{3}$ on $x[3n]$, $x[3n + 1]$, $x[3n + 2]$, and combine into $X[k]$, $X[k + \frac{N}{3}]$, $X[k + \frac{2N}{3}]$.
- Can generalize to the **radix- R** case ($N = R^M$).
- Can generalize to

$$N = \underbrace{N_1 \cdot N_2 \cdot N_3 \cdots}_{\text{product of prime numbers}}$$

(**mixed radix**): radix- N_1 stage $\rightarrow$ radix- N_2 stage $\rightarrow \cdots$

5 Applications

5.1 Convolution

Let $x[n]$, $0 \leq n \leq N - 1$, and $h[n]$, $0 \leq n \leq N - 1$. Direct linear convolution:

$$y[n] = \sum_{k=0}^{N-1} x[k] h[n - k], \quad 0 \leq n \leq 2N - 2.$$

N multiplications per $y[n] \Rightarrow 2N^2 - 2N$ multiplications.⁴

7 5.2 Circular Convolution Approach

Using the zero-padding result of Lecture 8 (pad to length $2N - 1$):

- 2 length- $(2N - 1)$ FFTs: $2[(2N - 1) \log_2(2N - 1)]$ mult.

⁴Bounding each of the $2N - 1$ outputs by N products gives $N(2N - 1) = 2N^2 - N$. Counting only the nonzero terms, every product $x[k]h[m]$ with $0 \leq k, m \leq N - 1$ appears exactly once, so the exact count is N^2 . Either way it is $\mathcal{O}(N^2)$.

- Pointwise multiplication: $(2N - 1)$ mult.
 - 1 length- $(2N - 1)$ inverse FFT: $(2N - 1) \log_2(2N - 1)$ mult.
- $\Rightarrow 3(2N - 1) \log_2(2N - 1) + (2N - 1)$ mult vs. $\mathcal{O}(N^2)$ directly.

In practice one pads to the next power of two $\geq 2N - 1$ so that the radix-2 FFT applies.

5.3 Frequency Responses

Recall

$$X[k] = X(e^{j\omega}) \Big|_{\omega=\frac{2\pi k}{N}}, \quad k = 0, \dots, N - 1, \quad H(e^{j\omega}) = \frac{\sum_{\ell=0}^M p_{\ell} e^{-j\omega\ell}}{\sum_{\ell=0}^N d_{\ell} e^{-j\omega\ell}}.$$

Evaluating $H(e^{j\omega})$ on a grid.

Let $S \geq \max(M, N)$ and zero-pad the coefficient sequences $\{p_{\ell}\}$, $\{d_{\ell}\}$ to length S .⁵ Then

$$H(e^{j\omega}) \Big|_{\omega=\frac{2\pi k}{S}} = \frac{\text{FFT}_S\{p\}[k]}{\text{FFT}_S\{d\}[k]}, \quad k = 0, \dots, S - 1.$$

8 6 Zero-Padding

Consider a length- N sequence $x[n]$ and “zero-pad” it to length $L > N$:

$$\tilde{x}[n] = \begin{cases} x[n], & 0 \leq n \leq N - 1, \\ 0, & N \leq n \leq L - 1. \end{cases}$$

⁵The coefficient sequences have lengths $M + 1$ and $N + 1$, so strictly $S \geq \max(M, N) + 1$ is needed for the length- S DFT to contain every coefficient. (This is what MATLAB’s `freqz` does internally.)

If we let

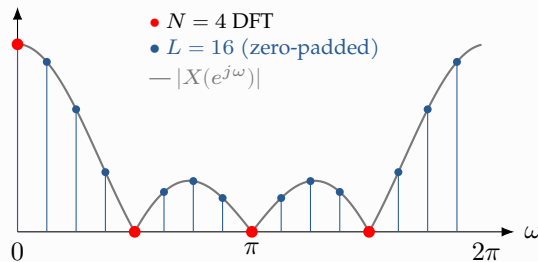
$$\tilde{X}[k] = \sum_{n=0}^{L-1} \tilde{x}[n] e^{-j\frac{2\pi kn}{L}} = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi kn}{L}} = X(e^{j\omega}) \Big|_{\omega=\frac{2\pi k}{L}}, \quad k = 0, \dots, L-1.$$

Property.

Zero-padding in discrete time gives **interpolation** but **does not** provide additional information = **periodic sinc interpolation** in the frequency domain.

Example (illustration, not in orig lecture note).

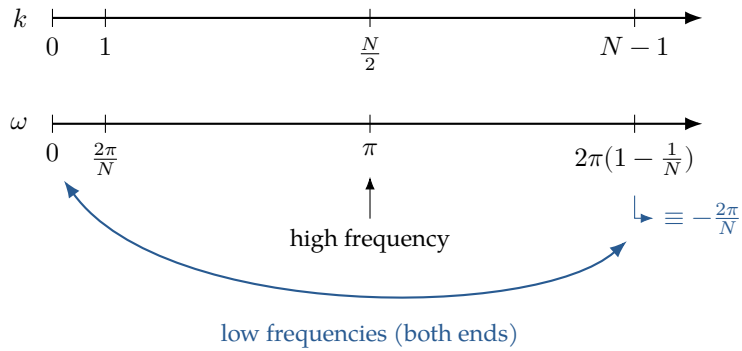
$x[n] = 1$ for $0 \leq n \leq 3$ ($N = 4$), so $|X(e^{j\omega})| = |\sin(2\omega)/\sin(\omega/2)|$ (Lecture 5). The 4-point DFT samples the DTFT only at $\omega = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, giving $\{4, 0, 0, 0\}$ — the lobes are invisible. Zero-padding to $L = 16$ samples the *same* curve more densely:



Both sets of samples lie on the same DTFT; zero-padding only reveals more of it (no new resolution).

9 7 DFT Index vs. Frequency

$$X[k] = X(e^{j\omega}) \Big|_{\omega=\frac{2\pi k}{N}}, \quad k = 0, \dots, N-1.$$



Note.

The `fftshift` command in MATLAB will place the **zero frequency at the center** instead of at the edge, i.e., it reorders $k = \frac{N}{2}, \dots, N-1, 0, \dots, \frac{N}{2}-1$ so that the axis runs over $-\pi \leq \omega < \pi$.

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 11.3–11.5
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 7